

Customer Support Ticket Analysis Prompt

The prompt below instructs an AI assistant to analyze each customer support conversation transcript in three steps: (1) evaluate conversation quality on several metrics, (2) classify the ticket by topic and intent, and (3) identify underlying root causes. It then outputs all results in a structured JSON format. The model is guided to use a chain-of-thought approach (breaking down the analysis step-by-step) but **only the final JSON** is returned as the answer.

Prompt (Chain-of-Thought Style)

You are given a customer support conversation transcript. Perform the following analysis:

1. Conversation Quality Assessment:

- Empathy & Personalization: Did the agent acknowledge the customer's emotions or needs? Assign a level (none/low/medium/high), quote a short evidence snippet from the conversation, and give a confidence score (0.0-1.0).
- Context Awareness: Did the agent appropriately reference prior messages? Assign level, evidence, confidence.
- Resolution Clarity: Are next steps or resolutions clear to the customer? Assign level, evidence, confidence.
- Accuracy & Correctness: Are facts, instructions, or policies stated correctly? Assign level, evidence, confidence.
- Efficiency & Tone: Is the response concise, polite, and on-brand? Assign level, evidence, confidence.

2. Topic and Intent Classification: Identify the main **topic** of the ticket (choose from predefined topics or propose a new one, setting ``is_new_topic`` true if new) and determine the customer's **intent** (e.g. asking a question, filing a complaint, requesting a refund, etc.).

3. Root Cause Identification: Determine the underlying reason for the customer's issue or escalation (from predefined root causes or suggest a new one, setting ``is_new_root_cause`` true if new).

4. Resolution Status: Indicate whether the issue was resolved by the end of the conversation (``is_resolved``: true or false).

After your analysis, output a JSON object with the following keys:

- ``quality_attributes``: an object containing the five metrics above, each with ``level``, ``evidence``, and ``confidence``.
- ``topic``: the chosen topic label.
- ``is_new_topic``: true/false flag for new topics.
- ``intent``: the identified intent label.
- ``root_cause``: the chosen root cause.

- `is\_new\_root\_cause`: true/false flag for a new root cause.
- `is\_resolved`: true/false for resolution status.

Each quality attribute must use level `none|low|medium|high`. Provide only the final JSON (no extra text).

Example output format:

```
```json
{
 "quality_attributes": {
 "empathy": {"level": "", "evidence": "", "confidence": 0.0},
 "context_awareness": {"level": "", "evidence": "", "confidence": 0.0},
 "resolution_clarity": {"level": "", "evidence": "", "confidence": 0.0},
 "accuracy": {"level": "", "evidence": "", "confidence": 0.0},
 "efficiency_tone": {"level": "", "evidence": "", "confidence": 0.0}
 },
 "topic": "",
 "is_new_topic": false,
 "intent": "",
 "root_cause": "",
 "is_new_root_cause": false,
 "is_resolved": false
}
```

## Sample Prompts and Outputs

### Example 1: Shipping Delay Issue

#### Transcript:

Customer: Hi, I'm upset that my order still hasn't arrived.  
Agent: I'm sorry to hear that. Let me check your order status.  
Customer: I paid for express shipping; it's been two weeks now.  
Agent: According to the tracking, the package was delayed by the courier.  
Agent: I will arrange a rush delivery so you get it by tomorrow.  
Customer: Thank you, I appreciate it.

#### Output JSON:

```
{
 "quality_attributes": {
 "empathy": {"level": "medium", "evidence": "I'm sorry to hear that.",
"confidence":0.85},
 "context_awareness": {"level": "high", "evidence": "According to the
tracking, the package was delayed by the courier.", "confidence":0.9},
 "resolution_clarity": {"level": "high", "evidence": "I will arrange a rush
delivery so you get it by tomorrow.", "confidence":0.9},
 "accuracy": {"level": "high", "evidence": "The agent's information matches
the order tracking details.", "confidence":0.9},
```

```

 "efficiency_tone": {"level": "high",
"evidence": "Responses are concise and polite.", "confidence": 0.9}
 },
 "topic": "Shipping",
 "is_new_topic": false,
 "intent": "Package_Delay",
 "root_cause": "Late Delivery",
 "is_new_root_cause": false,
 "is_resolved": true
}

```

## Example 2: New Topic / Website Bug

(Assume predefined topics are [Billing, Technical Issue, Account, Shipping, Returns].)

### Transcript:

Customer: I'm really frustrated. Your new site update has broken the search function - nothing shows up when I try to find products.  
 Agent: I apologize for the inconvenience. We have a ticket open for this bug.  
 Agent: Our developers are working on a fix and expect it to be resolved in 2 days.

### Output JSON:

```

{
 "quality_attributes": {
 "empathy": {"level": "medium", "evidence": "I apologize for the inconvenience.", "confidence": 0.8},
 "context_awareness": {"level": "high", "evidence": "We have a ticket open for this bug.", "confidence": 0.9},
 "resolution_clarity": {"level": "high", "evidence": "The issue will be fixed in 2 days.", "confidence": 0.9},
 "accuracy": {"level": "high", "evidence": "Information aligns with the known issue.", "confidence": 0.9},
 "efficiency_tone": {"level": "high", "evidence": "The agent is concise and polite.", "confidence": 0.9}
 },
 "topic": "Website Search Bug",
 "is_new_topic": true,
 "intent": "ReportIssue",
 "root_cause": "Software Bug",
 "is_new_root_cause": true,
 "is_resolved": false
}

```

(Here the issue does not match any predefined topic or root cause, so new labels are created and flagged.)

### Example 3: Multiple Tickets (Shipping & Billing)

#### Transcript 1:

Customer: Hi, I'm upset that my order still hasn't arrived.  
Agent: I'm sorry to hear that. Let me check your order status.  
Customer: I paid for express shipping; it's been two weeks now.  
Agent: According to the tracking, the package was delayed by the courier.  
Agent: I will arrange a rush delivery so you get it by tomorrow.  
Customer: Thank you, I appreciate it.

#### Transcript 2:

Customer: I noticed I was charged twice for my recent order.  
Agent: I apologize for the confusion. I will reverse the duplicate charge right away.  
Customer: Thank you, I appreciate it.

#### Output JSON:

```
[
 {
 "quality_attributes": {
 "empathy": {"level": "medium", "evidence": "I'm sorry to hear that.", "confidence": 0.85},
 "context_awareness": {"level": "high", "evidence": "According to the tracking, the package was delayed by the courier.", "confidence": 0.9},
 "resolution_clarity": {"level": "high", "evidence": "I will arrange a rush delivery so you get it by tomorrow.", "confidence": 0.9},
 "accuracy": {"level": "high", "evidence": "The agent's information matches the order tracking details.", "confidence": 0.9},
 "efficiency_tone": {"level": "high", "evidence": "Responses are concise and polite.", "confidence": 0.9}
 },
 "topic": "Shipping",
 "is_new_topic": false,
 "intent": "Package_Delay",
 "root_cause": "Late Delivery",
 "is_new_root_cause": false,
 "is_resolved": true
 },
 {
 "quality_attributes": {
 "empathy": {"level": "medium", "evidence": "I apologize for the confusion.", "confidence": 0.85},
 "context_awareness": {"level": "high", "evidence": "The agent identified the duplicate charge immediately.", "confidence": 0.9},
 "resolution_clarity": {"level": "high", "evidence": "The agent is reversing the duplicate charge now.", "confidence": 0.9}
 }
 }
]
```

```

 "accuracy": {"level": "high", "evidence": "The agent correctly addressed
the billing issue.", "confidence": 0.9},
 "efficiency_tone": {"level": "high",
"evidence": "The responses are brief and courteous.", "confidence": 0.9}
 },
 "topic": "Billing",
 "is_new_topic": false,
 "intent": "RefundRequest",
 "root_cause": "Billing Error",
 "is_new_root_cause": false,
 "is_resolved": true
 }
]

```

(Here we processed two conversation records at once, so the output is a JSON array with one object per transcript.)

#### Example 4: Single-Instruction Prompt (Account Login)

##### Prompt:

Analyze the following conversation transcript and output a JSON summary with the quality metrics, topic, intent, root cause, and resolution status as specified above.

##### Transcript:

```

Customer: I can't log into my account after the app update.
Agent: I apologize for the inconvenience. Please reset your password by
clicking 'Forgot Password'.
Customer: That fixed it, thanks.

```

##### Output JSON:

```

{
 "quality_attributes": {
 "empathy": {"level": "medium", "evidence": "I apologize for the
inconvenience.", "confidence": 0.8},
 "context_awareness": {"level": "high", "evidence": "The agent acknowledges
the login issue due to the update.", "confidence": 0.9},
 "resolution_clarity": {"level": "high",
"evidence": "The agent instructs to reset the password clearly.",
"confidence": 0.9},
 "accuracy": {"level": "high", "evidence": "The solution provided is
appropriate for the issue.", "confidence": 0.9},
 "efficiency_tone": {"level": "high", "evidence": "The response was concise
and polite.", "confidence": 0.9}
 },
 "topic": "Account Access",
 "is_new_topic": false,
 "intent": "Password_Reset",

```

```
"root_cause": "Software Bug",
"is_new_root_cause": true,
"is_resolved": true
}
```

These examples demonstrate how the prompt should be structured and the expected JSON output format for different types of customer support conversations. The actual conversation transcript and predefined topic/intent lists can be substituted as needed.

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